Machine Learning

Lecture 1: Introduction, Past and Present

Prof. Udit Bhatia

Civil Engineering; Computer Science and Engineering

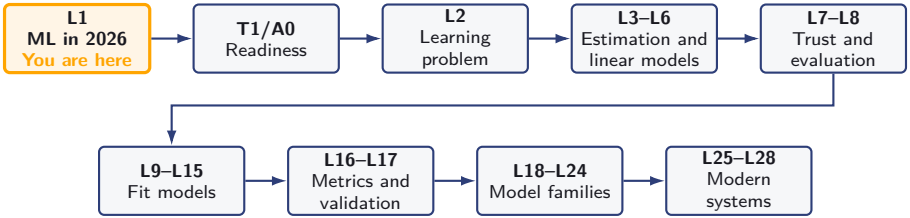
Prof. Krishna Prasad Miyapuram

Cognitive and Brain Sciences; Artificial Intelligence

Semester I, 2026



Course Roadmap: You Are Here



Lecture 1 motivates the whole course: what ML looks like in 2026, why the fundamentals still matter, and how ML results can mislead.

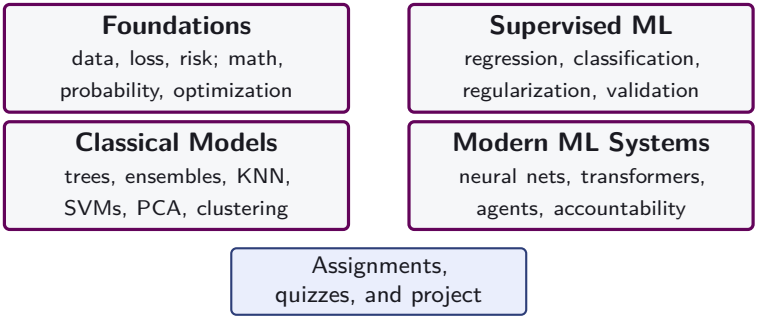
Course Coordinates

Item	Details
Course codes	ES 335 Machine Learning (UG) / ES 678 Machine Learning (PG)
Instructor	Prof. Udit Bhatia
Tutor	Prof. Krishna Prasad Miyapuram
Lectures	J1 Tuesday 14:00–15:20 and K2 Thursday 14:00–15:20, room 10/103
Tutorial	B2 Friday 08:30–09:50, Jagdish Patel Learning Theatre / Harsha Patel Learning Theatre

Summary

Bring the dedicated course notebook to every lecture and tutorial. Schedule or room changes, if any, will be announced through the official course channel.

Course Content



Summary

The course is organized around one spine: how to learn from data, evaluate honestly, and reason about deployment.

Lecture Objectives

By the end of Lecture 1, you should be able to:

1. explain why ML in 2026 still depends on classical foundations;
2. describe the ML loop from data to representation to model to decision;
3. recognize four ways ML results can mislead: aggregation, metrics, shortcuts, and model capacity;
4. state the basic learning notation: $\mathcal{S}_{\text{train}}$, $\boldsymbol{x}$, y, f, ℓ, R, and $\widehat{R}$;
5. understand the course expectations for assignments, AI tools, quizzes, and classroom conduct.

Guiding Question

When a model looks impressive, what would make the result untrustworthy?

The Claim For This Course

In 2026, ML is not just "fit a model".

It is the craft of building **data-driven systems** whose predictions, explanations, and decisions remain reliable outside the notebook.

Insight

The tools will change quickly. The fundamentals of data, loss, risk, generalization, and evaluation will stay useful.

ML In 2026

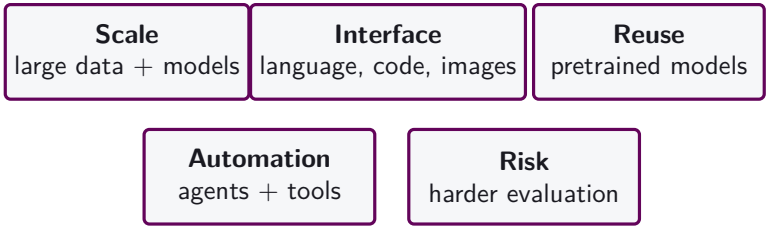
A Short History Of The Interface



Summary

The interface moved from hand-coded rules to learned functions to large pretrained systems. The question "what has been learned from data?" remains central.

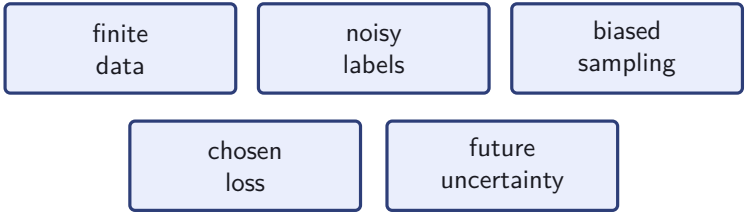
What Changed



Warning

Powerful interfaces make it easier to produce outputs. They do not make the outputs trustworthy by default.

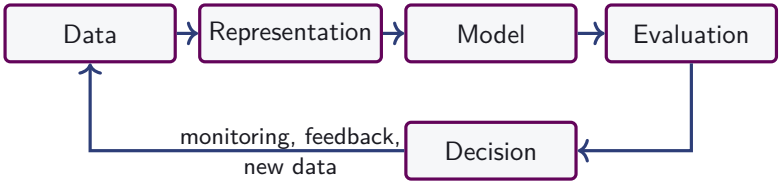
What Did Not Change



Insight

ML remains the problem of making reliable claims about unseen cases from limited observations.

The 2026 ML Loop



Summary

This course studies the whole loop. The model is only one part of the system.

Four Paradoxes

Why Start With Paradoxes?

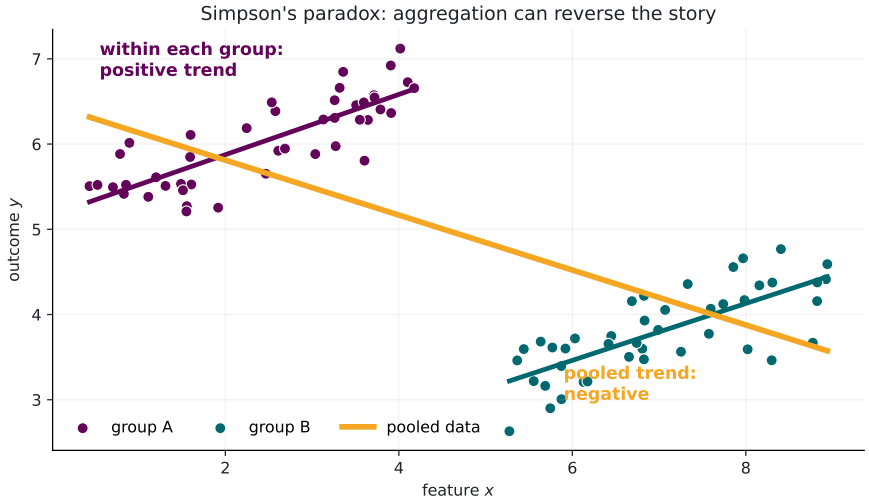
ML is powerful partly because it finds patterns.

ML is dangerous partly because it finds **the wrong patterns** with great confidence.

Question

The first skill in ML is not coding a model. It is noticing when the apparent pattern is not the claim you think it is.

Paradox 1: Aggregation Can Reverse A Conclusion



What Simpson's Paradox Teaches

Conditioning

Before trusting a relationship between X and Y, ask whether a third variable G changes the story.

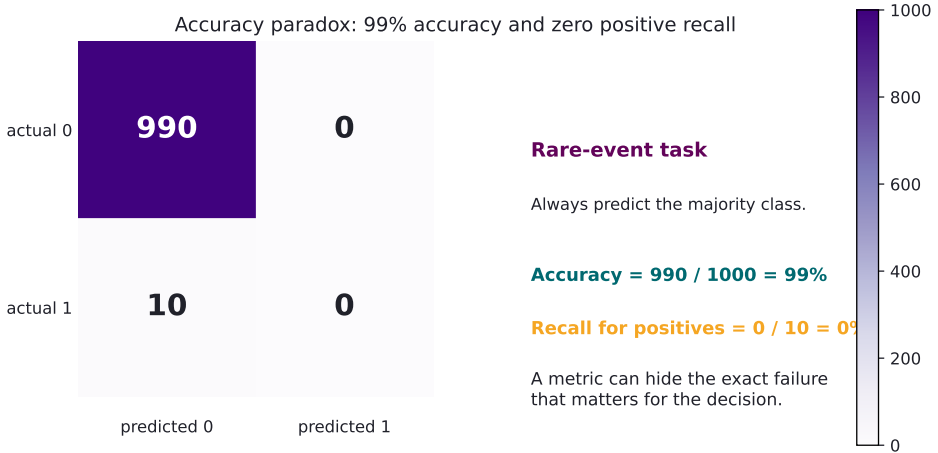
In ML language:

$$p(Y \mid X) \quad \text{may look very different from} \quad p(Y \mid X, G).$$

Insight

Subgroups, context, and sampling mechanisms are not details. They can change the sign of the conclusion.

Paradox 2: High Accuracy Can Be Useless



Rare-event task

Always predict the majority class.

Accuracy = 990 / 1000 = 99%

Recall for positives = 0 / 10 = 0%

A metric can hide the exact failure that matters for the decision.

What The Accuracy Paradox Teaches

For rare-event tasks, the majority class can dominate accuracy.

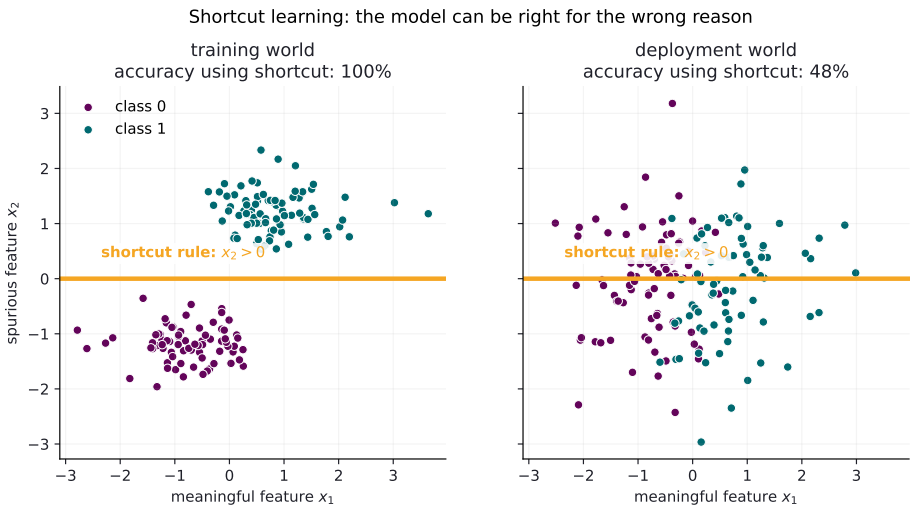
Examples:

- flood damage detection
- disease screening
- fraud detection
- equipment failure prediction

Warning

The metric must match the decision. Otherwise the best score may reward the wrong behavior.

Paradox 3: Right Prediction, Wrong Reason



What Shortcut Learning Teaches

Models exploit whatever predictive signal is easiest.

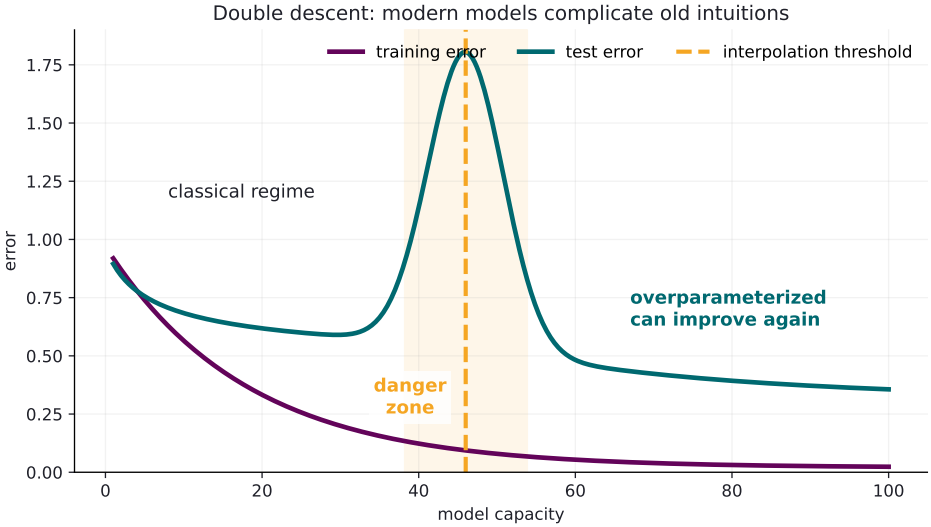
That signal may be a real mechanism, or it may be a shortcut:

background instead of object	hospital instead of disease
sensor artifact instead of event	future information instead of present information

Question

Would this feature still be available, valid, and causal enough at deployment time?

Paradox 4: Bigger Models Can Behave Strangely



What Double Descent Teaches

Classical intuition says: more flexible models overfit.

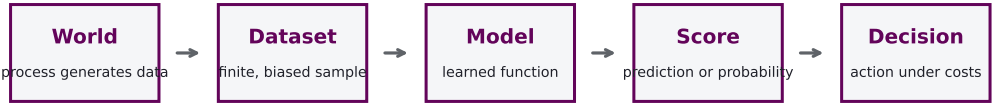
Modern ML adds a twist: after the interpolation threshold, very large models can sometimes improve again.

Insight

This does not remove the need for validation. It makes careful validation more important.

The Learning Language

From Data-Generating Process To Decision



Every arrow hides assumptions. ML is partly the art of making them explicit.

The Basic Learning Problem

Learning Problem

Given examples

$$\mathcal{S}_{\text{train}} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^{n},$$

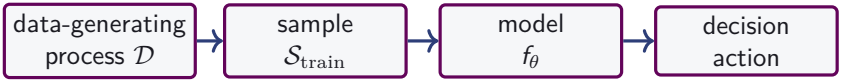
learn a function

$$f : \mathcal{X} \to \mathcal{Y}$$

that performs well on future examples.

$\boldsymbol{x}_i$: input $\quad$ y_i: target $\quad$ f: model

Four Objects We Must Not Confuse



Summary

Many failures come from confusing one object for another: the sampled dataset for the data-generating process, the score for the decision, or the fitted model for understanding.

Loss: What Error Means

A loss function scores one prediction:

$$\ell(f(\boldsymbol{x}), y).$$

Task	Example loss
Regression	$(\hat{y} - y)^2$
Classification	$\mathbb{1}\{\hat{y} \neq y\}$
Probabilistic classification	$-\log \hat{p}_y$

Warning

Choosing the loss is choosing what kind of mistake matters.